Socio-Cognitive Privacy Guardian

EYE-RIS | AI-BASED PRIVACY ROBOT | AUTONOMOUS SMART CAR

An intelligent mobile robot that enforces privacy using AI perception, gesture recognition, and autonomous navigation — built for homes, offices, and healthcare environments.

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Objective

Build a mobile robot that ensures privacy through AI detection, physical camera actuation, and fully autonomous navigation — with zero false positives.

Key Use Cases

- **Homes** — prevents unwanted surveillance
- **Offices** — protects sensitive data zones
- **Healthcare** — ensures patient confidentiality

System Architecture & Methodology

EYE-RIS integrates a multi-layer cognitive pipeline — from human detection through gesture validation to physical privacy enforcement and autonomous movement.

Human Detected

YOLO on Raspberry Pi webcam

Gesture Recognition

Unlock or no-gesture triggers privacy

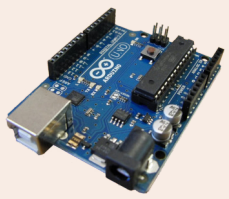
Cognitive Decision

Validate intent and send servo signal

Enforce Privacy

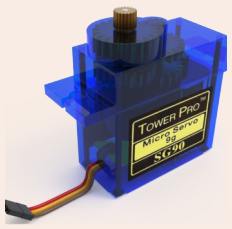
Camera tilts down; smart car navigates

Privacy actions are triggered only after multi-layer cognitive validation — ensuring no false positives and no unwarranted exposure.



Arduino UNO + L298N Motor Driver

Drives motor speed via PWM on pins 5–6 and direction logic on pins 7–11. Receives drive signals from the main perception Arduino.



Servo Motor

Connected to pin 10 of the main Arduino UNO. Physically tilts the camera downward when a human is detected — the core privacy actuation mechanism.

Ultrasonic Sensor

Provides real-time obstacle detection, feeding distance data into the autonomous navigation decision loop.

Smart Car Chassis

Mobile platform with a dedicated Arduino for motor control. Receives commands from the main UNO, enabling full autonomous traversal of the environment.